

Multi-Layer DoS Detection and Severity Scoring System Using Hybrid ML and DL Approaches

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
CAIDA	Center for Applied Internet Data Analysis
CNN	Convolutional Neural Network
CSP	Cloud Service Provider
CUSUM	Cumulative Sum
DL	Deep Learning
DSR	Design Science Research
EWMA	Exponentially Weighted Moving Average
FP	False Positive
GCP	Google Cloud Platform
ICMP	Internet Control Message Protocol
IDS	Intrusion Detection System
IP	Internet Protocol
IPS	Intrusion Prevention System
ISP	Internet Service Provider
L3 / L4	Layer 3 (Network) / Layer 4 (Transport)
LSTM	Long Short-Term Memory
ML	Machine Learning
MLP	Multilayer Perceptron
MSSP	Managed Security Service Provider
MTTR	Mean Time To Respond
PCAP	Packet Capture
RF	Random Forest
SDG	Sustainable Development Goal (United Nations)
SLA	Service Level Agreement
SME	Small and Medium Enterprise
SVM	Support Vector Machine
SYN	Synchronize (TCP handshake flag)
TDFA	Time-Dependent Finite Automata
TP	True Positive
UNB	University of New Brunswick
XDP	eXpress Data Path

ABSTRACT

The proposed severity-aware decision support mechanism enables Security Operations Centre (SOC) analysts and network administrators to prioritize incident responses based on quantified risk scores rather than binary alerts, reducing Mean Time to Respond (MTTR) and supporting risk-informed mitigation decisions. This research contributes to the IAS (Information Assurance & Security) research cluster at SLIIT and aligns with UN Sustainable Development Goals SDG 9 and SDG 16.

The hybrid detection framework employs Random Forest (RF) and Support Vector Machine (SVM) classifiers for structured feature-based classification, complemented by Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks for capturing temporal traffic patterns and sequential behavioral signatures. Evaluation is conducted using the CICDDoS2019 benchmark dataset (Canadian Institute for Cybersecurity, University of New Brunswick) and the CAIDA DDoS 2007 dataset. Target performance metrics include a minimum detection accuracy of 90% and F1-scores of at least 0.85 per component.

The system targets four distinct DoS attack vectors: SYN Flood (TCP, Layer 4), UDP Flood (UDP, Layer 4), IP Fragmentation Attack (Layer 3), and ICMP Flood (ICMP, Layer 3). Each component independently performs feature extraction, hybrid Machine Learning and Deep Learning (ML/DL) classification, packet quantification, and weighted severity computation using predefined impact weights: SYN Flood (3.0), UDP Flood (2.5), Fragmentation Attack (2.0), and ICMP Flood (1.5). A Final Severity Aggregator consolidates individual component scores into a Combined Scoring Ranking Table, mapping the aggregated severity score to one of four severity levels: Low, Medium, High, or Critical.

Denial of Service (DoS) attacks continue to pose a critical threat to modern digital infrastructure, targeting Layer 3 and Layer 4 protocols through high-volume packet floods and connection-state exhaustion techniques. Existing detection systems predominantly operate on binary classification — distinguishing attack traffic from normal traffic — without quantifying the relative severity or impact priority of concurrent multi-vector attacks. This research proposes a Multi-Layer DoS Detection and Severity Scoring System that addresses this gap through a four-component parallel detection framework integrated with a centralized Severity Scoring Engine.

Keywords: Denial of Service, DoS Detection, Severity Scoring, Machine Learning, Deep Learning, SYN Flood, ICMP Flood, UDP Flood, IP Fragmentation, CNN, LSTM, Random Forest, SVM, CICDDoS2019, Network Security, SOC, IDS

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1. INTRODUCTION

1.1. Background and Context

Denial of Service (DoS) attacks represent one of the most persistent and damaging cyber threats facing modern digital infrastructure. By overwhelming target systems with malicious traffic, DoS attacks render critical services unavailable to legitimate users. Organizations including enterprises, cloud service providers, Internet Service Providers (ISPs), financial institutions, healthcare systems, and government networks are primary victims due to their dependence on continuous network availability. According to recent global statistics, the frequency and sophistication of DoS and DDoS attacks have increased dramatically; in 2023 alone, major cloud providers reported thousands of multi-vector attacks per week, with individual incidents peaking at traffic volumes exceeding 71 million requests per second [1].

The impact of DoS attacks spans multiple dimensions. Technically, unresolved DoS attacks cause service outages, performance degradation, and resource exhaustion across Layer 3 (Network Layer) and Layer 4 (Transport Layer) protocols. Economically, prolonged downtime results in direct financial losses, Service Level Agreement (SLA) violations, and significant reputational damage. Socially, disruptions in critical services such as healthcare, banking, and e-commerce erode public trust. From a legal and ethical perspective, failure to detect and respond to high-severity attacks may lead to regulatory non-compliance and negligence claims, particularly in critical infrastructure sectors [2].

Current DoS detection systems largely operate on binary classification — determining whether traffic is an attack or normal — without differentiating the relative severity of individual attack types. Network administrators and Security Operations Centre (SOC) analysts require an intelligent system that not only detects multi-layer DoS attacks but also quantifies their real-world impact to support fast and informed decision-making [1] [2]. This research addresses that gap by proposing a severity-aware, multi-layer DoS detection framework built on hybrid Machine Learning (ML) and Deep Learning (DL) approaches.

This project aligns with the IAS (Information Assurance & Security) research cluster at SLIIT, focusing on securing digital infrastructures through intelligent and adaptive security mechanisms. It also contributes to two United Nations Sustainable Development Goals: SDG 9 (Industry, Innovation, and Infrastructure) by strengthening resilient digital infrastructure, and SDG 16 (Peace, Justice, and Strong Institutions) by enhancing cyber resilience against disruptive attacks [5].

1.2. Literature Review

The body of existing research on DoS detection can be broadly categorized into three generations: rule-based and statistical approaches, machine learning-based classification systems, and recent hybrid and deep learning architectures.

1.2.1 Rule-Based and Statistical Approaches

Early DoS detection mechanisms relied on anomaly detection algorithms applied to network traffic statistics. Bogdanoski et al. (2013) conducted a comprehensive analysis of SYN Flood attacks using OPNET simulation, demonstrating that during a SYN flood, TCP segment delay increases twofold and voice packet delay variation escalates by tenfold compared to baseline [1]. The paper introduces two anomaly detection algorithms: the Adaptive Threshold algorithm, which monitors SYN packet counts against an Exponentially Weighted Moving Average (EWMA) baseline, and the CUSUM (Cumulative Sum) algorithm based on change-point detection. These methods are effective against high-intensity attacks but degrade significantly for low-intensity, slow-rate attacks.

Manna and Amphawan (2012) provided a comprehensive taxonomy of SYN flooding detection mechanisms, classifying them into three categories: router data structure-based schemes (using Bloom Filters and Counting Bloom Filters), statistical analysis-based schemes, and Artificial Intelligence-based schemes using fuzzy logic and neural networks [3]. Their performance comparison reveals that all categories generate notable false positive rates and struggle with high CPU overhead. Notably, AI-based methods such as fuzzy logic systems outperform statistical methods in low-false-negative rate but require careful threshold calibration.

Kumar and Selvakumar (2011) proposed a distributed approach to detect flooding-based DoS attacks using a statistical mechanism that analyzes IP header fields, specifically source IP entropy, to distinguish attack traffic from legitimate traffic [4]. Their method demonstrated effectiveness in identifying high-rate UDP and ICMP floods with minimal false positives under controlled conditions. However, the system assumed a static traffic baseline, making it less adaptive to dynamic network environments.

1.2.2 Machine Learning-Based Approaches

ML-based detection systems have shown significant improvement over rule-based approaches. Syafiuddin et al. (2023) proposed an AdaBoost ensemble classifier for detecting SYN Flood and UDP Lag attacks using the CICDDoS2019 dataset [5]. Their system, incorporating chi-square feature selection and 10-fold cross-validation, achieved 94.5% accuracy, 74.1% precision, and 67.3% recall. This demonstrates that ensemble methods offer a strong balance between detection

accuracy and computational feasibility for real-time deployment. The study also confirms CICDDoS2019 as a reliable benchmark dataset covering multiple DoS attack types.

Commercial and open-source tools such as Suricata IDS, HAProxy, and Nginx are widely deployed for traffic-based mitigation. Romadhon et al. (2024) compared HAProxy and Nginx for mitigating UDP flood attacks, finding HAProxy superior due to its event-driven architecture and lower resource consumption under high-volume conditions [5]. However, these systems perform binary detection or simple threshold filtering, with no mechanism for attack severity differentiation.

Aamir and Zaidi (2019) evaluated multiple supervised learning classifiers — including Random Forest, Decision Tree, Naïve Bayes, and k-Nearest Neighbor — on DDoS traffic datasets and found that Random Forest consistently outperformed other models with an accuracy exceeding 99% on balanced datasets [6]. Their work highlighted the critical role of feature selection in reducing computational overhead and emphasized that class imbalance in real-world traffic datasets significantly degrades detection recall for minority attack classes. This finding directly informs the feature engineering strategies adopted in this research.

1.2.3 Deep Learning and Hybrid Architectures

Dimolianis et al. (2021) proposed a signature-based SYN flood detection and mitigation system combining supervised ML models (Random Forest and Multilayer Perceptron) with programmable XDP data-plane firewalls [7]. Their approach achieved a True Negative Rate above 99% across all test scenarios and reduced malicious signature sets by 99.99% through a multi-objective optimization process. The MLP model demonstrated superior generalization across different attack variants, while Random Forest was more interpretable. This work validates the effectiveness of ML-based signature classification for real-time environments but lacks any severity assessment layer.

AlDossary et al. (2023) proposed a Time-Dependent Finite Automata (TDFA) model for SYN Flood detection, implementing a three-state system (**Safe** → **Critical** → **DoS Attack**) that triggers when more than 10 SYN requests arrive within a 10-second window. While the TDFA approach is computationally lightweight and deterministic, it relies on a static threshold and cannot adapt to varying attack intensities or classify severity levels. Deep learning approaches using LSTM, CNN, TCN, and hybrid CNN–LSTM models have further enhanced detection performance for high-speed networks [2]. Multi-layered defense architecture proposed by Sutradhar and Sarkar (2025) detect attacks across network, transport, and application layers, improving resilience against multi-vector DoS attacks [7].

Garg et al. (2022) proposed a hybrid CNN-LSTM model specifically designed for detecting volumetric DoS attacks in IoT-enabled network environments [8]. By leveraging CNN layers for spatial feature extraction from packet-level data and LSTM layers for modeling temporal traffic sequences, their architecture achieved an F1-score of 0.96 on the CICDDoS2019 dataset. Importantly, their work demonstrated that temporal modeling through LSTM significantly improves detection of slow-rate and low-volume attack patterns that evade purely feature-based classifiers — a key motivation for incorporating CNN and LSTM within each component of the proposed system.

Despite the above advances, a consistent gap exists: existing research primarily treats DoS detection as a binary or multi-class classification problem without quantifying attack severity or relative impact. Most systems do not differentiate the relative danger of attack types such as SYN flood versus ICMP flood, limiting effective response prioritization. Additionally, many architectures focus either on detection or mitigation, lacking an integrated severity-aware decision mechanism.

1.3 Research Gap

The key limitation identified across the reviewed literature is the absence of severity quantification in DoS detection systems. Current systems, regardless of their detection mechanism (statistical, ML, or DL), produce outputs that classify traffic as either attack or normal, or at best identify the attack type. They do not provide any measure of how dangerous or impactful a detected attack is relative to others occurring simultaneously.

This is a critical shortcoming in real operational environments. A SOC analyst simultaneously receiving alerts for an ICMP Flood and a SYN Flood needs to know not only that both attacks are present, but which represents the greater immediate threat to infrastructure. Existing tools do not provide this contextual severity ranking. Furthermore, no existing system integrates multi-layer detection (covering ICMP, TCP SYN, UDP, and fragmentation attack vectors simultaneously) with a weighted impact scoring mechanism.

This research addresses this gap by introducing a severity-aware scoring engine that assigns impact-based weights to different attack types based on their known protocol-level impact and resource exhaustion potential: SYN Flood (3.0), UDP Flood (2.5), Fragmentation Attack (2.0), and ICMP Flood (1.5). This scoring is integrated with a hybrid ML/DL detection framework, enabling risk-prioritized mitigation decisions that are currently absent from the literature.

2. OBJECTIVES

2.1 Main Objective

To design, implement, and evaluate a SYN Flood detection and high-impact severity scoring module — as the highest-weighted component (Weight = 3.0) of a multi-layer DoS detection framework — that accurately identifies TCP SYN Flood attacks using hybrid Machine Learning and Deep Learning models, and quantifies attack severity to support risk-aware, prioritized mitigation decisions by SOC analysts and network administrators.

2.2 Specific Objectives

1. To extract and analyses relevant SYN Flood traffic features including SYN packet rate, SYN/FIN ratio imbalance, half-open connection count, inter-arrival time between SYN requests, source IP entropy, and TCP flag distribution from network traffic datasets (CICDDoS2019 and CAIDA).
2. To implement and evaluate hybrid ML models (Random Forest and SVM) for structured TCP SYN traffic feature classification, achieving a target detection accuracy of at least 90%.
3. To implement and evaluate deep learning models (CNN or LSTM) for capturing temporal SYN Flood traffic patterns and improving detection performance under high-volume and low-rate attack conditions.
4. To develop and validate a weighted severity scoring engine that multiplies the detected SYN attack packet count and half-open connection count by an impact weight of 3.0 and maps the resulting score to severity levels: Low, Medium, High, and Critical.
5. To generate a structured SYN Flood attack severity ranking table that enables SOC analysts to assess and prioritize SYN flood attack responses based on computed severity scores and connection-exhaustion impact.
6. To validate the severity scoring mechanism against attack impact indicators including traffic volume, persistence duration, TCP backlog queue exhaustion rate, and system resource utilization metrics.

3. METHODOLOGY

3.1 Research Approach

This research adopts a Design Science Research (DSR) methodology, which is appropriate for creating and evaluating innovative IT artifacts. The research follows an experimental and iterative development cycle: **(1) problem identification and motivation**, **(2) design and development of**

the detection artifact, (3) experimental evaluation using benchmark datasets, and (4) results communication. The project is structured as a prototype-level near real-time framework, suitable for experimental evaluation rather than full production deployment.

3.2 Technical Methods and Algorithms

3.2.1 Traffic Capture and Feature Extraction

Network traffic is monitored and Layer 3/Layer 4 features are extracted using time-window analysis. For SYN Flood detection specifically, relevant features include SYN packet count per time window, SYN/FIN ratio imbalance, half-open connection count, inter-arrival time between SYN packets, source IP diversity (entropy), TCP flag distribution, packet rate, and flow duration. These features are derived from established literature [1] [9] and confirmed as effective discriminators for SYN Flood traffic classification.

3.2.2 Multi-Layer Attack Detection — Hybrid ML/DL Framework

A hybrid detection approach is adopted combining Machine Learning and Deep Learning models. Machine Learning models — specifically Random Forest and Support Vector Machine (SVM) are selected for their interpretability, computational efficiency, and proven performance on structured, tabular network traffic features. These models operate on statistical feature vectors extracted from time windows.

Deep Learning models — specifically Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks — are incorporated to capture temporal traffic patterns and sequential behavioral signatures that are not captured by static ML models. The CNN layers extract local feature patterns from packet sequences, while LSTM captures long-range temporal dependencies in traffic flow behavior. The exact model configuration will be refined during empirical experimentation.

3.2.3 Attack Quantification

Detected attack packets are counted separately per attack type within each time window: ICMP packet count, SYN packet count and connection attempt count, UDP packet burst count, and fragmented IP packet count.

3.2.4 Severity Scoring Engine — Core Contribution

The Severity Scoring Engine computes a weighted impact score for each detected attack type. Each packet count is multiplied by a predefined impact-based weight:

Attack Type	Weight	Layer	Protocol
SYN Flood	3.0	Layer 4 (Transport)	TCP
UDP Flood	2.5	Layer 4 (Transport)	UDP
Fragmentation Attack	2.0	Layer 3 (Network)	IP
ICMP Flood	1.5	Layer 3 (Network)	ICMP

Table 1Severity Scoring

SYN Flood carries the highest weight (3.0) due to its severe impact on TCP connection-state exhaustion, as demonstrated by simulation results showing a sevenfold increase in TCP segment delay under DoS-SYN conditions [1]. Weights are initial estimates derived from existing literature and will be refined empirically during the project.

3.2.5 Severity Classification

The total aggregated severity score is mapped to four severity levels: Low (score < threshold_1), Medium (threshold_1 ≤ score < threshold_2), High (threshold_2 ≤ score < threshold_3), and Critical (score ≥ threshold_3). Threshold values will be empirically determined during evaluation.

3.3 Tools and Platforms

Python 3.10+ will be the primary implementation language, leveraging scikit-learn for Random Forest and SVM models, TensorFlow/Keras for CNN and LSTM architectures, and Pandas/NumPy for data preprocessing and feature engineering. Wireshark and tcpdump will be used for packet-level traffic analysis and feature extraction. Jupyter Notebooks will support iterative experimentation and model evaluation. CICDDoS2019 Dataset – Primary benchmark dataset containing labelled ICMP flood, SYN flood, UDP flood, and fragmentation attack traffic.

3.4 Data Collection and Ethical Considerations

This project uses publicly available benchmark datasets: CICDDoS2019 (Canadian Institute for Cybersecurity, University of New Brunswick) and CAIDA datasets. CICDDoS2019 contains millions of labeled network traffic records covering ICMP flood, SYN flood, UDP flood, and fragmentation attacks, validated in recent literature [5]. No human participants, personal data, or

real-world attack execution against live systems are involved. This research strictly adheres to defensive cybersecurity ethics and SLIIT institutional research policies.

3.5 Validation Strategy and Performance Metrics

The severity scoring framework will be validated by comparing computed severity levels against attack impact indicators including traffic volume, persistence duration, and system resource utilization. ML/DL model performance will be evaluated using Accuracy, Precision, Recall, F1-Score, and True Positive / False Positive Rates. Where possible, results will be cross-validated against expert-defined severity expectations from existing DoS literature. Performance will be benchmarked against baseline binary detection systems identified in the literature review.

3. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Overview

The proposed system is designed as a four-component parallel detection framework converging into a centralized severity aggregation engine. The architecture follows an Input–Process–Output model operating on near real-time or offline network traffic.

4.2 Component Interactions and Data Flow

Network traffic (Layer 3/Layer 4 packets) enters the system simultaneously and is routed to four parallel detection components based on protocol type. Each component independently performs: **(1) packet counting**, **(2) feature extraction**, **(3) ML/DL-based attack detection**, **(4) severity scoring via the weighted engine**, and **(5) severity classification**. Each component produces an individual scoring table. All four scoring tables are then fed into the Final Severity Aggregator, which generates a Combined Scoring Ranking Table providing the overall severity level.

The data flow proceeds as follows: **Raw Traffic** → **Protocol Filtering** → **Per-Component Detection** → **Individual Severity Scores** → **Aggregator** → **Combined Severity Ranking** → **SOC Alert Output**. The output includes: detected attack types, individual severity scores per attack, total aggregated severity score, and the overall severity level (Low/Medium/High/Critical).

4.3 Architecture Components Description

Stage	Component	Description
INPUT	Network Traffic Source	CICDDoS2019 / CAIDA datasets; near real-time or offline PCAP traffic containing TCP protocol packets
INPUT	Feature Extraction Module	Extracts SYN Flood-specific L4 features using time-window statistical analysis: SYN packet rate, SYN/FIN ratio, half-open connection count, inter-arrival time between SYN requests, source IP entropy, and TCP flag distribution
PROCESS	ML Detection Engine	Random Forest and SVM classifiers for structured TCP SYN traffic feature classification
PROCESS	DL Detection Engine	CNN/LSTM networks for capturing temporal SYN burst patterns and sequential TCP connection-state behaviors
PROCESS	SYN Packet Counter	Counts detected malicious SYN packets and half-open connection attempts per time window
PROCESS	Severity Scoring Engine	Applies the highest impact weight of 3.0 to the SYN packet and connection count; computes the SYN severity score reflecting TCP connection-state exhaustion potential
PROCESS	Severity Classifier	Maps computed SYN severity score to severity levels: Low / Medium / High / Critical
OUTPUT	ICMP Scoring Table	Structured table with attack type, count, score, severity level
OUTPUT	Final Aggregator Feed	Feeds severity data to the Combined Scoring Ranking Table

Table 2 Architecture Process Description

4.4 Scalability and Security Considerations

The system is designed as a modular prototype; each detection component can be independently scaled or replaced without affecting the aggregator. The parallel processing architecture allows future horizontal scaling to accommodate higher traffic volumes. From a security standpoint, the system operates in a read-only, passive monitoring mode — it does not inject packets into the network or modify traffic flows, ensuring no interference with legitimate operations. All evaluations are conducted on anonymized, publicly available datasets, preventing exposure of sensitive network data.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders for this system are: Network Administrators (responsible for network infrastructure monitoring and maintenance), SOC Analysts (Security Operations Centre analysts who respond to security alerts in real time), IT Security Managers (who require risk-informed dashboards for decision-making), and Research Community (academics and security researchers who evaluate detection frameworks). Secondary stakeholders include cloud service providers, ISPs, financial institutions, and healthcare IT departments that depend on continuous network availability.

5.2 Functional Requirements

- ✚ FR1: The system shall monitor network traffic at Layer 3 and Layer 4 and extract relevant packet features per time window.
- ✚ FR2: The system shall detect ICMP Flood, SYN Flood, UDP Flood, and IP Fragmentation attacks independently using hybrid ML/DL models.
- ✚ FR3: The system shall count detected attack packets per type per time window.
- ✚ FR4: The system shall apply predefined impact-based severity weights to each attack type and compute individual severity scores.
- ✚ FR5: The system shall aggregate individual severity scores into a total severity score and map it to a severity level (Low, Medium, High, Critical).
- ✚ FR6: The system shall generate a Combined Scoring Ranking Table listing all detected attacks with their individual and total severity scores.
- ✚ FR7: The system shall process near real-time or offline network traffic as input.
- ✚ FR8: The system shall support both CICDDoS2019 and CAIDA datasets for experimental evaluation.

5.3 Non-Functional Requirements

Security

The system operates in passive monitoring mode without modifying network traffic. All datasets used are publicly available and anonymized. No real-world attacks are executed.

Performance

The system shall achieve a minimum overall detection accuracy of 90% on benchmark datasets. Individual component F1-scores shall meet or exceed 0.85. Severity score computation shall complete within 500 milliseconds per time window for prototype evaluation.

Usability

System outputs shall be presented in a human-readable format (scoring tables and severity levels) understandable to SOC analysts without requiring deep technical expertise. Results shall include clear labels indicating attack type, individual score, and overall severity level.

Scalability

The modular component architecture shall support addition of new attack-type detection modules without redesigning the aggregation layer.

Reliability

The system shall maintain consistent detection performance across different attack volumes, durations, and packet rates, as validated across multiple dataset subsets.

Maintainability

Each detection component shall be implemented as an independently testable module with clearly documented interfaces to the Final Severity Aggregator.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market includes medium-to-large enterprise IT departments, cloud service providers (CSPs), managed security service providers (MSSPs), Internet Service Providers (ISPs), financial services institutions, and healthcare IT systems. The primary customer persona is a SOC Analyst or Network Security Engineer at an organization operating 24/7 network infrastructure who currently lacks automated severity prioritization in their threat detection pipeline.

6.2 Value Proposition

Existing DoS detection tools provide binary alerts — they tell operators that an attack is happening but not how urgent the response needs to be. This system's unique value proposition is severity-aware alerting: it provides not just attack detection but a quantified risk score that enables automated response prioritization. This reduces Mean Time To Respond (MTTR) and enables more efficient allocation of SOC analyst attention.

6.3 Revenue Model and Pricing Strategy

The commercialization pathway begins as an open-source research prototype to build academic credibility and community validation. Subsequently, a SaaS licensing model could be developed

offering tiered pricing: a Basic tier for SMEs providing standard DoS detection and severity scoring, and an Enterprise tier with custom severity weight tuning, real-time dashboard integration, and API access for SIEM integration. Revenue could also be generated through consulting services for customizing severity weights to specific industry verticals.

6.4 Competitive Advantage

The key competitive differentiator is the severity scoring engine, which is absent from existing commercial IDS/IPS solutions. While products like Suricata, Snort, and commercial SIEM platforms detect DoS attacks, none provide an integrated multi-attack severity ranking mechanism. This gap represents a meaningful market opportunity, particularly for organizations subject to SLA guarantees where rapid severity-based response is a contractual obligation.

6.5 Intellectual Property Considerations

The severity scoring algorithm, including the impact-based weighting methodology and the aggregation framework, represents potential intellectual property that could be protected through a software patent or trade secret upon empirical refinement and commercial maturity. The research outputs will be published in IEEE or ACM venues, establishing prior art and academic credibility.

7. BUDGET AND JUSTIFICATION

✓ **Total Estimated Budget: LKR 35,000**

Item	Cost (LKR)	Category	Justification
High-performance laptop / workstation upgrade	0	Hardware	Required for running DL model training (CNN/LSTM) locally
Cloud compute (AWS/GCP) GPU instances	25,000	Cloud Services	GPU acceleration required for CNN/LSTM model training on large datasets
Cloud storage for dataset hosting	5,000	Cloud Services	CICDDoS2019 and CAIDA datasets require persistent cloud storage
Python libraries & dev tools (open-source)	0	Software	scikit-learn, TensorFlow, Keras, Pandas — all free/open-source
Human effort — 4 members × 20 weeks	0	Human Effort	Allocated stipend contribution for research assistantship activities

Miscellaneous (printing, meetings, travel)	5,000	Miscellaneous	Document printing, team coordination, external review meetings
Total		LKR 35,000	

Table 3 Budget

8. WORK BREAKDOWN STRUCTURE

ID	Member	Task	Duration	Deliverable
T1	Farhan R	Literature review on SYN Flood detection, TCP connection-state exhaustion, and hybrid ML/DL approaches	Weeks 1–2	Annotated bibliography
T2	Farhan R	Dataset acquisition and preprocessing — isolating SYN Flood traffic records from CICDDoS2019 and CAIDA	Week 2	Approved TAF + Proposal
T3	Farhan R	SYN Flood feature extraction and selection — SYN packet rate, SYN/FIN ratio, half-open connection count, inter-arrival time, source IP entropy, TCP flag distribution	Weeks 3–6	ICMP detection module v1
T4	Farhan R	Random Forest and SVM model implementation for TCP SYN traffic classification	Weeks 3–6	SYN detection module v1
T5	Farhan R	CNN/LSTM deep learning model implementation for temporal SYN burst pattern detection	Weeks 3–6	Fragmentation detection module v1
T6	Farhan R	SYN packet counting and half-open connection tracking module development	Weeks 3–6	UDP detection module v1
T7	Farhan R	Severity scoring engine implementation (Weight = 3.0) applied to SYN packet count and connection attempts	Weeks 7–8	Scoring engine prototype
T8	Farhan R	Severity classification and SYN Flood ranking table generation (Low / Medium / High / Critical)	Weeks 9–10	Integrated prototype
T9	Farhan R	Model evaluation and validation against CICDDoS2019 ground truth and attack impact indicators	Weeks 11–13	Evaluation results report
T10	Farhan R	Integration of SYN component output with Final Severity Aggregator	Weeks 12–13	Validation report

T11	All team members	Final documentation and report writing for SYN Flood detection component	Week 14	Progress report
T13	Farhan R	Final report writing and documentation	Weeks 17–18	Final report draft

Table 4 Work Breakdown Structure

9. GANTT CHART

The following Gantt chart provides a semester-wise breakdown of all tasks, milestones, and deliverables across project timeline.

Task	W1-2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Literature Review & Setup	■	■	■										
Dataset Acquisition & Preprocessing	■	■	■										
SYN Feature Extraction & Selection			■	■	■								
Random Forest & SVM Implementation				■	■	■	■						
CNN/LSTM Model Implementation						■	■	■					
SYN Packet Counter Module							■	■	■				
Severity Scoring Engine (Weight=3.0)								■	■	■			
Severity Classification & Ranking Table										■	■	■	
Evaluation & Validation												■	■
Integration with Final Aggregator												■	■
Final Documentation & Report Writing												■	■

Table 5 Gant Chart

■ = Active task period | Milestones: Week 3 (Literature Complete), Week 14 (Progress Review), Week 18 (Final Submission)

References

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11. APPENDICES

Appendix A — Risk Analysis

Risk	Likelihood	Impact	Mitigation Strategy
Dataset unavailability / corruption	Low	High	Use multiple backup datasets (CICDDoS2019 + CAIDA); maintain local copies
ML model underfitting / poor accuracy	Medium	High	Iterative hyperparameter tuning; try multiple algorithms; increase feature set
Insufficient compute resources for DL training	Medium	Medium	Use Google Colab Pro or cloud GPU instances; optimize batch size
Scope creep — severity weight refinement consuming excessive time	Medium	Medium	Fix initial weights as estimates; document refinement as future work
Team member availability disruption	Low	Medium	Maintain shared codebase (GitHub); document all components thoroughly
Severity scoring not validated against real attack impact	Medium	High	Use CICDDoS2019 ground truth labels and traffic volume metrics as proxy validators

Table 6 Risk Analysis

Appendix B — Ethical Considerations Summary

This project does not require formal ethical clearance as it meets all of the following criteria: (1) No human participants are involved in any experimental procedures. (2) No personal, sensitive, or identifiable information is collected or processed. (3) All experiments are conducted using synthetic or publicly available anonymized datasets (CICDDoS2019, CAIDA). (4) No real-world attack execution against live systems is performed at any stage of the research. (5) The research focuses exclusively on defensive cybersecurity mechanisms. The project complies with SLIIT institutional research policies and the ethical principles of responsible data usage, transparency in methodology, and non-maleficence.

Appendix C — Dataset Summary

Dataset	Source	Attack Types Covered	Size (approx.)
CICDDoS2019	Canadian Institute for Cybersecurity, UNB	SYN Flood, UDP Flood, ICMP Flood, Fragmentation, and 10+ DDoS variants	~50M labeled records
CAIDA DDoS 2007	Center for Applied Internet Data Analysis	Backbone-level DDoS traffic traces	~21GB pcap

Table 7 Data Summary